Intergenerational Mortgage Cosigning: Credit Access and Mortgage Pricing

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Motivation, Definition, Mechanism

Motivation (US)

- Equity is the main financial asset for many households; mortgages are the key lever to homeownership.
- In 2023, about 92% of new home purchases in the US used a mortgage.
- The US mortgage market is \$12.59 trillion ($\approx$ 70% of consumer debt; $\approx$ 50% of GDP) (2023).

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Definition: What is a cosigner?

- A cosigner is not the main borrower, but is **legally responsible for repayment if the borrower defaults**.
- Cosigners **do not obtain ownership rights** in the property.

Mechanism and Research Question

Key mechanism

- Relaxes children's mortgage constraint **without a down-payment transfer**, but shifts default risk onto parents.

Research question

- How does intergenerational cosigning shape the **housing distribution**?
- What are the consequences for **intergenerational wealth transmission**?

Why should we care?

- **Wealth inequality:** homeownership is the main channel for wealth accumulation.
- **Housing affordability:** cosigning may relax credit constraints for young households.
- **Financial stability:** reallocates default risk to family and potentially affects leverage.

Preview of Results

Empirics:

1. Intergenerationally cosigned mortgages account for about 3% of mortgages in the US ($7.4 billion p.a.).
2. Children use cosigning to financial limitations in the housing market.
3. Cosigning parents are relatively liquidity constraint.
4. Natural Experiment: Cosigning eases access t the mortgage market but not conditions of contracts.

Theory:

1. Quantitative model where cosigning relaxes PTI constraints leading to reallocating housing access and risk across generations.
2. The model quantifies distributional effects on **homeownership, leverage, default, and bequests**.

Outline

Literature

Literature

- **Portfolio Choice with Housing**: Cocco 2005, Eichenbaum, Rebelo, and Wong n.d., Mian and Sufi 2011-08, Mian, Rao, and Sufi 2013, Mian and Sufi 2014, Mian, Sufi, and Trebbi 2015

$\Longrightarrow$ Cosigning shapes housing distribution

- **Intergenerational Wealth Transfers**: Black et al. 2022, De Nardi 2004, De Nardi and Fella 2017, Druedahl and Martinello 2022, Koltikoff and Summers 1981, Nekoei and Seim 2023, Modigliani 1988, Ohlsson, Roine, and Waldenström 2020, Saez and Zucman 2016,

$\Longrightarrow$ Cosigning affects intergenerational wealth transmission

- **Parental Support and Housing Affordability**: Allen et al. 2024 and Benetton, Kudlyak, and Mondragon 2024

$\Longrightarrow$ Document and quantify cosigning in the US

Predictions

A two-period OLG model following Allen et al. 2024 with liquid and housing wealth where

- Agents can cosign at the end of the first period and get a warm glow from doing so.
- Young when cosigning, pay when old.
- Young when cash decision, pay when young.

yields the following predictions:

1. Cosigning lowers rejection rates
2. People to live in pricier areas and/or
3. Cosigning expands housing demand and/or
4. Lower mortgage interest rate

5. Cosigning parents have less liquid assets than parents with cash transfer

Set Up

Data

Data Sources

Table 1: Key Features of HMDA, Web-Scraped Data and PSID

Feature	HMDA
Scope	Near-universe of **loan-level applications** (2018–2024)
Key Info	**Mortgage, Borrower, Lender** details; Application outcomes (originated/rejected)
Cosigning ID	Via age structure ($\geq$ 19-year age difference)
Drawback	Only first two mortgagors listed

Data Sources

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Scope	Near-universe of **loan-level applications** (2018–2024)	Universe of **originated mortgages** in a given county (currently 30)
Key Info	**Mortgage, Borrower, Lender** details; Application outcomes (originated/rejected)	**Mortgagors, Mortgagees, Loan Amount, Geographic Area, Origination Date**
Cosigning ID	Via age structure ($\geq$ 19-year age difference)	Via more than two mortgagors listed
Drawback	Only first two mortgagors listed	No age information

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Feature	HMDA	Web-Scraped Data	PSID
Scope	Near-universe of **loan-level applications** (2018–2024)	Universe of **originated mortgages** in a given county (currently 30)	**Household data** with parental links (2019–2025, 3 waves)
Key Info	**Mortgage, Borrower, Lender** details; Application outcomes (originated/rejected)	**Mortgagors, Mortgagees, Loan Amount, Geographic Area, Origination Date**	**Household portfolio** (Net Wealth); Family Linkages.
Cosigning ID	Via age structure ($\geq$ 19-year age difference)	Via more than two mortgagors listed	
Drawback	Only first two mortgagors listed	No age information	

Descriptives

HMDA - Evidence on Cosigning

- 2.6% of mortgage applications are intergenerationally cosigned. (7.4 billion USD p.a.)
- With webscraped data: 5.47%

Table 2: By Age

Age Bin	Cosigned (%)	w/ Webscraping
<25	9.57	12.36
25–34	2.99	5.60
35–44	1.30	4.04

Table 3: By Year

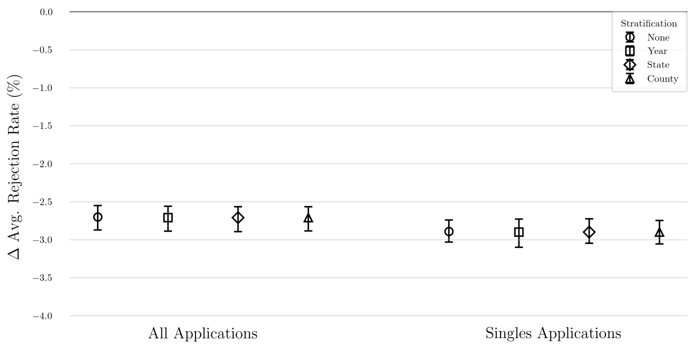
Year	Cosigned (%)	w/ Webscraping
2018	2.55	5.41
2019	2.12	4.37
2020	2.21	5.04
2021	2.41	5.16
2022	2.88	5.55
2023	3.11	5.64
2024	3.19	7.01

States

Prediction 1 - Application Rejection Rates

Prediction 1: Cosigners face lower rejection rates

Figure 1: Mean Difference in Application Rejection Rates



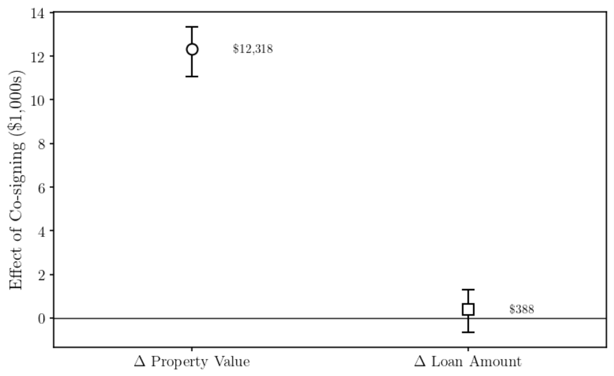
Unweighted Average across Property Value x Loan Amount cell.

Prediction 2 - Housing Demand

Conditional on Acceptance

Prediction 2: Cosigners have higher housing demand

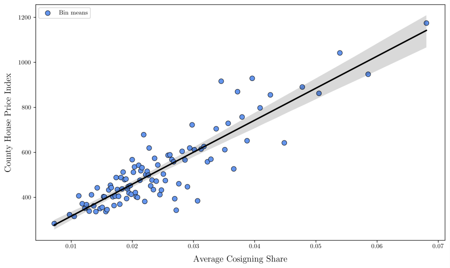
Figure 2: Mean Difference in Property Value and Loan Amount



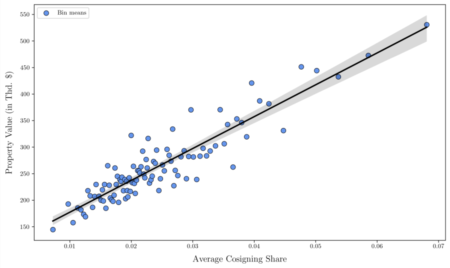
Prediction 3 - House Prices

Prediction 3.: Cosigned Mortgagees live in pricier areas

County House Price Index



Avg. Property Value

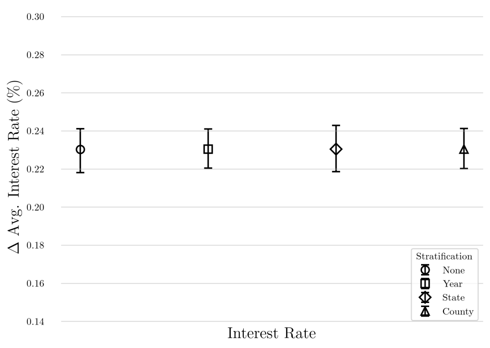


House Price Index Larger Counties

Prediction 4 - Interest Rate

Prediction 4: Cosigners face lower interest rates

Figure 3: Mean Difference in Interest Rates



Unweighted Average across Property Value x Loan Amount cell.

Robustness of Correlations

Table 4: Parental Cosigning: Extensive and Intensive Margins

	Prediction 1 **Rejection Rate**
Parental cosigning	-0.0154***
	(0.0017)
Lender FE	Yes
State FE	Yes
Year FE	Yes
Demographics	Yes
Property Value & Loan Amount	Yes
R^2	0.009
Observations	6,672,649

Rejection Rates, Interest Rates, Prop. Value, Loan Amount

Robustness of Correlations

Table 4: Parental Cosigning: Extensive and Intensive Margins

	Prediction 1 **Rejection Rate**	**Prediction 2** **Log Property Value**
Parental cosigning	-0.0154***	0.0818***
	(0.0017)	(0.0027)
Lender FE	Yes	Yes
State FE	Yes	Yes
Year FE	Yes	Yes
Demographics	Yes	Yes
Property Value & Loan Amount	Yes	—
R^2	0.009	0.180
Observations	6,672,649	5,795,691

Rejection Rates, Interest Rates, Prop. Value, Loan Amount

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Table 4: Parental Cosigning: Extensive and Intensive Margins

	Prediction 1 Rejection Rate	Prediction 2 Log Property Value	Prediction 2 Log Loan Amount
Parental cosigning	-0.0154***	0.0818***	0.0510***
	(0.0017)	(0.0027)	(0.0030)
Lender FE	Yes	Yes	Yes
State FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Demographics	Yes	Yes	Yes
Property Value & Loan Amount	Yes	—	—
R^2	0.009	0.180	0.163
Observations	6,672,649	5,795,691	5,795,691

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	Prediction 1 **Rejection Rate**	**Prediction 2** **Log Property Value**	**Prediction 2** **Log Loan Amount**	**Prediction 4** **Interest Rate**
Parental cosigning	-0.0154***	0.0818***	0.0510***	-0.0161
	(0.0017)	(0.0027)	(0.0030)	(0.0103)
Lender FE	Yes	Yes	Yes	Yes
State FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Demographics	Yes	Yes	Yes	Yes
Property Value & Loan Amount	Yes	—	—	Yes
R^2	0.009	0.180	0.163	0.000
Observations	6,672,649	5,795,691	5,795,691	5,790,238

Rejection Rates, Interest Rates, Prop. Value, Loan Amount

Intensive Margin - w/ webscraping

Table 5: Cosigning and Intensive Margin (Webscraping)

	Log Property Value
Parental cosigning	0.1200***
	(0.0063)
County FE	Yes
Lender FE	No
State FE	No
Year FE	Yes
Demographics	Yes
R^2	0.562
Observations	158,354

Intensive Margin - w/ webscraping

Table 5: Cosigning and Intensive Margin (Webscraping)

	Log Property Value	**Log Loan Amount**
Parental cosigning	0.1200***	0.0988***
	(0.0063)	(0.0066)
County FE	Yes	Yes
Lender FE	No	No
State FE	No	No
Year FE	Yes	Yes
Demographics	Yes	Yes
R^2	0.562	0.481
Observations	158,354	158,354

Intensive Margin - w/ webscraping

Table 5: Cosigning and Intensive Margin (Webscraping)

	Log Property Value	**Log Loan Amount**	**Interest Rate**
Parental cosigning	0.1200***	0.0988***	0.1045***
	(0.0063)	(0.0066)	(0.0175)
County FE	Yes	Yes	Yes
Lender FE	No	No	No
State FE	No	No	No
Year FE	Yes	Yes	Yes
Demographics	Yes	Yes	Yes
R^2	0.562	0.481	0.398
Observations	158,354	158,354	143,307

Connecting to PSID

- No direct connection from HMDA to PSID.
- Use Multiple Imputation Chain Equation (MICE) to impute cosigning in PSID
- Idea: Find N closest neighbors and match mean. Repeat N times for entire sample.
- Impute intergenerational linkages
- Predicted share of cosigned mortgages 7.48%

Who are the parents?

Prediction 5: Cosigning parents have less liquid wealth than helpers

Table 6: Who are the parents? - Some Means

	Savings*	Total Wealth	Income	Share Owner***	Years Mortgage	Years Educ.
Not Cosigner	67,304	997,398	118,239	73.6 (%)	7.14	13.85
Cosigner	48,481	926,186	128,885	69.8 (%)	7.39	13.85

Children

Summary and Interpretation

Key findings

- **Cosigning is economically meaningful:** 2.6% of applications in HMDA ($7.4bn p.a.); **5.5%** with webscraping.
- **Lower rejection rates** (Prediction 1): cosigned applications are **less likely rejected** (≈ -1.5pp).
- **Higher housing demand** (Prediction 2): cosigners have **higher** property values (≈ 10pp) and larger loans (≈ 8pp).
- **Pricier Areas** (Prediction 3): strong correlation with county level prices.
- **Lower interest rates** (Prediction 4): limited evidence on lower interest rates.
- **Liquidity-constraint parents** (Prediction 5): have lower liquid wealth than the average parent of first-time homebuyers.

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Endogeneity / selection concern

- **Self-selection:** households may decide to cosign exactly when the child is a **riskier borrower**.
- Results conflate effect of cosigning with selection bias.

  $\Longrightarrow$ Need for exogenous variation to isolate key margins of cosigning.

Natural Experiment

Institutional Framework - US

Primary mortgage market

- Households take out mortgages from banks, credit unions, and mortgage companies.
- Lenders screen and price loans.
- Rejections, interest rates, and cosigning are determined.

Secondary mortgage market

- Lenders sell eligible loans to Fannie Mae and Freddie Mac (GSEs).
- GSEs pool loans into MBS and guarantee investors against default.
- For this guarantee, GSEs charge guarantee fees (g-fees).

Reform in the Secondary Market

Reform (FHFA pricing change)

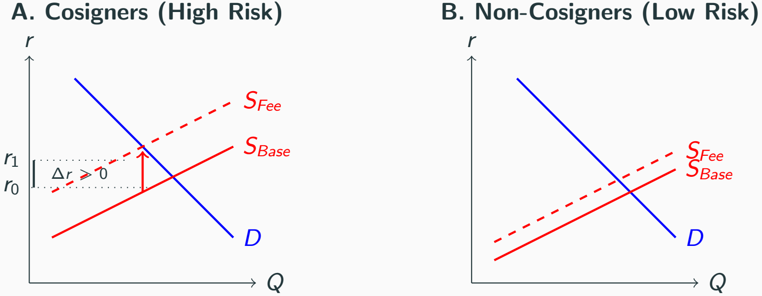
- Announced: January 5th, 2022. Implemented: April 1st, 2022.
- FHFA increased g-fees on high-balance loans (0.25 percent and 0.75 percent)
- Equal for Cosigned and Non-cosigned loans.
- Increase in marginal cost of banks for these loans.
- Interpretation: Exogenous cost shock at the conforming loan limit, which shifts banks' credit supply for loans just above the cutoff.

Idea:
Exploit heterogeneity in the pass-through of increased fee by cosigning status.
Banks perception of risk by cosigning status.

Exposure to GSE | Over time | Delinquency | Interest Rates

Contract-Specific Risk-Based Pass-Through

If Cosigners riskier: G Fee increase will be passed through to cosigners more.



The Setup: Potential Outcomes Framework

Notation:

- Let i denote the loan application, l the lender, and t the time period.
- **Running Variable ($\tilde{X}_{it}$):** Loan amount centered at the Conforming Loan Limit (CLL).
$$\tilde{X}_{it} = \text{LoanAmount}_{it} - \text{Limit}_t$$
- **Treatment (D_{it}):** Assignment to High-Cost (GSE Fee) regime.
$$D_{it} = \mathbb{1}(\tilde{X}_{it} > 0)$$
- **Outcomes (Y_{it}):** Interest Rate, Rejection Probability.

The Identification Problem: Endogenous Sorting

In a standard Regression Discontinuity (RD) design, identification requires continuity of potential outcomes at the cutoff.

The Violation:

- Borrowers have incentives to "bunch" below the limit ($\tilde{X}_{it} < 0$) to avoid fees.
- Sorting is **non-random**: wealthy borrowers manipulate X; riskier borrowers remain above $X > 0$.

Bias in Standard RD

Consequently, the standard Cross-Sectional RD estimator at time t yields:

$$\tau_{SRD,t} = \tau_{\text{True}} + \underbrace{\left[\lim_{x \downarrow 0} E[Y(0)|x] - \lim_{x \uparrow 0} E[Y(0)|x]\right]}_{\text{Selection Bias } B_t \neq 0}$$

Bias

Identification Strategy: Difference-in-Discontinuities

To solve the sorting problem, we exploit the time dimension at the threshold:

- **Pre-Period ($t = 0$):** Low penalty baseline.
- **Post-Period ($t = 1$):** High penalty regime.

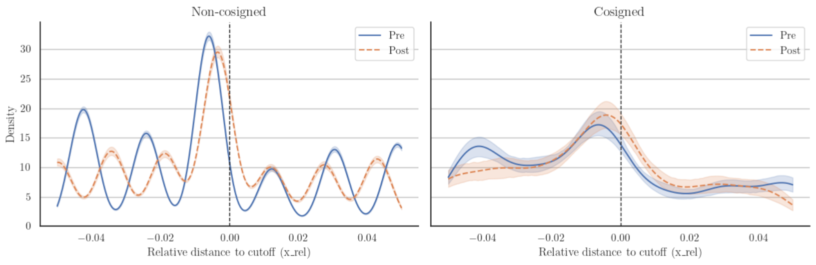
Key reference: **Difference-in-Discontinuities (Diff-in-Disc)** (Grembi, Nannicini, and Troiano, 2016).

Core Intuition: Relax the assumption that potential outcomes are continuous ($B_t = 0$) to assume the sorting is constant over time ($B_1 = B_0$).

$$\hat{\tau}_{DiDisc} = \tau_{SRD,t=1} - \tau_{SRD,t=0} = (\tau + B_1) - (0 + B_0) = \tau$$

Constant Bunching Over Time

Figure 4: Density Function for 2021 and 2022 by Cosigning Status



Testing Validity: Stability of Sorting Intensity

Formal Density Test (McCrary) Local linear model on binned log-densities (y_{bt}) where loans i are binned into bin b:

$$y_{bt}^{k} = \alpha^{k} + \delta_1^{k}\text{Post} + \delta_2^{k}\mathbb{1}_{>0} + \beta^{k}(\mathbb{1}_{>0} \times \text{Post}_t) + \epsilon_{bt}^{k} \qquad \forall k \in (\text{Cosigner, Non-Cosigner})$$

Null Hypothesis: $\beta = 0$
(The jump in density is constant over time).

Results (Change in Log-Jump):

Sample	$\hat{\beta}$ (SE)	p-val
Non-Cosigned	-4.57 (3.19)	0.17
Cosigned	-3.13 (2.85)	0.29

Conclusion: *Fail to reject the null. The intensity of manipulation did not change post-reform for either group.*

Econometric Specification

To allow for heterogeneous effects and distinct sorting mechanisms, we estimate the model **separately** for each subsample $k \in \{\text{Cosigner, Non-Cosigner}\}$:

$$Y_{ilt}^k = \alpha^k + \beta_1^k D_{it} + \beta_2^k \text{Post}_t + \boldsymbol{\beta_3}^k (D_{it} \times \text{Post}_t) + \lambda^k(\tilde{X}_{it}) + \mathbf{Z}_{it}\gamma^k + \delta_l^k + \epsilon_{ilt}^k$$

Definitions:

- $D_{it} = \mathbb{1}(\tilde{X}_{it} > 0)$: Above Conforming Limit indicator.
- $\beta_{\mathbf{3k}}$: **Parameter of Interest** (Causal Effect for group k).
- $\lambda_k(\cdot)$: Flexible linear slopes (allowed to vary by group k).
- δ_l: Lender Fixed Effects (clustered SEs).

Hypothesis Testing: After estimation, we test for the statistical difference between the groups:

$$H_0 : \beta_3^{\text{Cosigner}} - \beta_3^{\text{Non-Cosigner}} = 0$$

Main Results: Rejection Rates

Table 7: Rejection Rates Comparison

Group	Probability of Rejection: Pre-Reform ($T = 0$)	Probability of Rejection: Post-Reform ($T = 1$)	Decomposition of Change: Total Change (Post - Pre)	Decomposition of Change: Reform Effect (β_3)
Non-Cosigner	5.4%	7.5%	+2.1 pp	**+1.5 pp**
Cosigner	8.2%	13.4%	+5.2 pp	**-2.3 pp**
Difference	*2.8 pp*	**5.9 pp**	*3.1 pp*	**-3.8 pp**

Unconditional Mean: 7.57%

Main Results: Interest Rates

Table 8: Interest Rate Comparison

Group	Total Interest Rate Level Pre-Reform ($T = 0$)	 Post-Reform ($T = 1$)	Decomposition of Change Total Change (Post - Pre)	 Reform Effect (β_3)
Non-Cosigner	3.99%	5.04%	+1.05%	**-1.85%**
Cosigner	3.92%	5.65%	+1.73%	**-0.69%**
Difference	*+0.07*	**-0.61**	*-0.68*	**-1.16*** **

Unconditional Mean: 4.11%

Key Takeaway:

- **Pre-Reform:** Both groups have similar rates ($\approx$ 3.9%).
- **Post-Reform:** Non-Cosigners rise to 5%, while Cosigners remain higher at 5.6%.

Main Results: Discount Points

Table 9: Discount Points Comparison

	Total Discount Points Level		Decomposition of Change	
Group	**Pre-Reform** ($T = 0$)	**Post-Reform** ($T = 1$)	**Total Change** (Post - Pre)	**Reform Effect** (β_3)
Non-Cosigner	\$871	-\$5,896	-\$6,767	**-\$8,756****
Cosigner	\$892	+\$3,107	+\$2,215	**+\$216**
Difference	*-\$21*	-\$**9,003**	*-\$8,982*	-\$**8,972****

Unconditional Mean: \$1100 Bandwidth

Endogenous Sorting after Reform

Differential sorting on observables?, Endogenous subgroup selection?

Table 10: Borrower Composition Check

	Diff-in-Disc Estimate (β_3)		
Outcome	**Non-Cosigner**	**Cosigner**	**Difference**
Log Loan Amount	0.060***	0.048***	-0.012
	(0.00)	(0.00)	(0.46)
Log Property Value	0.046***	0.040**	-0.006
	(0.00)	(0.01)	(0.72)
Log Income	-0.020	0.116	0.137
	(0.39)	(0.23)	(0.17)
Unconditional Mean Log Loan Amount: 12.82			
Unconditional Mean Log property Value: 13.01			
Unconditional Mean Log Income: 11.56			
Change in Cosigning Probability (β_3^{Prob})			
-0.000*** (Unconditional Mean: 0.033)			

Estimates of β_3 using optimal bandwidths. Standard Errors are clustered at the lender level.

- **No Differential Sorting:** "Jumps" in Loan Amount, Property Value, and Income are statistically identical for both groups ($p > 0.10$).
- **No Sorting into Contracts:** Share of cosigned mortgage does not change.

Local Parallel Trends

Estimate the model at a "Placebo Cutoff" (80% of the actual limit) to test if the heterogeneous pricing arises naturally.

	Placebo Estimates ($\beta_3^{Placebo}$)		
Outcome	**Non-Cosigner**	**Cosigner**	**Difference**
Interest Rate (%)	0.563***	0.303*	**-0.260**
	(0.00)	(0.06)	(0.13)
Discount Points ($)	266	1,035	**769**
	(0.52)	(0.10)	(0.30)
App. Rejection	-0.023	-0.030	**-0.007**
	(0.16)	(0.24)	(0.83)

Table 11: Estimates at a placebo cutoff (80% of CLL)

- **No statistically significant difference** between Cosigners and Non-Cosigners at the placebo threshold.
- For Interest Rates, the placebo effect is positive, whereas the effect at the real cutoff was negative.

Summary and Interpretation

Key findings

- Cosigning works on the extensive margin and not on the intensive margin.
- Correlation differences in intensive margin driven by selection bias.
- Banks' pass-through of a common shock is equal across cosigning status on the extensive margin.
- Banks' pass-through of a common shock is larger for cosigners on the intensive margin.
- Consistent with interpretation that cosigned mortgages are riskier.

Quantitative Model

Model Environment and Timing

Partial equilibrium model with

Demographics and uncertainty

- Overlapping generations, lifecycle length $j = 1, \ldots, J$.
- Idiosyncratic, uninsurable income shocks $z_j \in \mathcal{Z}$ follow a Markov process $\Pi(z'|z)$.

Assets and housing

- Risk-free asset $a \geq 0$ with return r.
- Rental housing size $h^R \in \mathcal{H}^R$ at price p^R.
- Owned housing size $h \in \mathcal{H}^O$ at price p^O.

Life events

- In final period, agent can bequeath assets.
- At age $j_B + 30$ (parental help age): parents choose *no help* / *cosign* / *cash*.
- At age $J - 30$, agents receive an inheritance.
- At age j_B (buying age): household chooses rent vs. buy with mortgage.
- During mortgage term (M years): owners choose pay vs. default.

State Space (One Lifecycle Agent, Two Life Stages)

Same agent over the lifecycle: the "young/child" and "old/parent" problems are the **same individual's** decisions at different ages.

State Space

$$s \equiv (j, a, z, z^p, \omega, h, h^c, \eta)$$

- j = age, a = liquid assets, z = idiosyncratic productivity
- z^p = parent's type
- ω = housing status:

$$\omega \in \{RB,\ PD\}$$

- h = housing stock
- $h^c \in \{1, 2, 3\}$: realized help received when young (none/cosign/cash)
- $\eta \in \{1, 2, 3\}$ = help commitment made when old (none/cosign/cash)

Parental help received when young

When the agent is young and enters the housing market, the realized help regime is:

$$h^c \in \{1, 2, 3\}, \quad 1 = \text{none},\ 2 = \text{cosign},\ 3 = \text{cash}.$$

Before it is realized, the young agent forms beliefs

$$\pi^c(z^p) = \big(\Pr(h^c = 2 \mid z^p), \Pr(h^c = 3 \mid z^p)\big).$$

Recursive Objects (Same Agent, Stage-Dependent Problems)

Stage-dependent value functions of the same agent

$$V_{RB}(s), \qquad V_{PD}(s)$$

- $V_{RB}(s)$: renter/buyer stage (young ages, tenure choice)
- $V_{PD}(s)$: homeowner stage (mortgage repayment vs default)

Help decision when old (same agent later in life)
At age $j = j_P$, the agent becomes a parent and chooses

$$\eta \in \{1, 2, 3\} \quad \text{(none/cosign/cash)}$$

which determines the help regime faced by their own child in the next generation:

$$\eta \longrightarrow h^c \ \text{ for the child.}$$

Interpretation

- The only difference between "young" and "old" is **the age index** j (and hence which constraints/choices are active).
- Intergenerational linkage enters through z^p and η.

Young Stage: Rent vs. Buy

At the buying age $j = 15$, with i.i.d. Type-I extreme value taste shocks $\varepsilon^R, \varepsilon^B$ scaled by σ_{RB}:

$$V_{RB}(s) = \mathbb{E}_\varepsilon\Big[\max\big\{V_{RB}^{rent}(s)+\sigma_{RB}\varepsilon^R,\ V_{RB}^{buy}(s)+\sigma_{RB}\varepsilon^B\big\}\Big], \qquad s = (j, a, z, z^p, \omega, h^c, \eta).$$

Equivalently (log-sum):

$$V_{RB}(s) = \sigma_{RB}\log\left(\exp\left(\frac{V_{RB}^{rent}(s)}{\sigma_{RB}}\right) + \exp\left(\frac{V_{RB}^{buy}(s)}{\sigma_{RB}}\right)\right).$$

Buying

$$V_{RB}^{buy}(s) = \max_{a',\, h\in\mathcal{H}^O}\big\{u(c,h,1) + \beta\,\mathbb{E}\left[V_{PD}(s')\right]\big\}$$

subject to

$$c = we(j,z) + (1+r)a - a' + \underbrace{\left(\lambda_m p^O h - p^O h\right)}_{\substack{\text{downpayment}\\ =-(1-\lambda_m)p^O h}} + \underbrace{T(z^p)}_{\text{parental cash transfer if } h^c=3}.$$

Renter Default

Mortgage Access and Payment Construction

Mortgage contract (purchase of house size h at price p^O)

$$\text{House value} = p^O h, \qquad L = p^O h - (1-\lambda_m) p^O h$$

- L = loan principal
- $\text{DP}(h^c)$ depends on help regime: none / cosign / cash

Fixed-rate annuity payment (maturity M)

$$m(L, R(z), M) = \frac{R(z)(1+R(z))^M}{(1+R(z))^M - 1} \cdot L.$$

Mortgage access (reduced form PTI screening)

$$\text{PTI}(h, h^c, z, z^p) \equiv \frac{m(L, M)}{w\, e(j_B, z) \ + \ \mathbf{1}\{h^c = 2\}\, w\, e(j_P, z^p)}.$$

$$\text{Approved} \iff \text{PTI}(h, h^c, z, z^p) \leq \overline{\text{PTI}}.$$

Key link: cosigning affects only access decision.

Old Stage: Irreversible Help Commitment (η)

At age $j = j_P$, the parent makes a **one-time, irreversible** help commitment.

Parent problem (commitment choice with logit shocks)

$$V_P(j_P, a, z, \omega, h, \eta = 1) = \max_{\eta \in \{1,2,3\}} \left\{ \tilde{V}_P(j_P, a, z, \omega, h; \eta) + \varepsilon^{\eta} \right\},$$

where $\varepsilon^{\eta'}$ are i.i.d. Type-I Extreme Value (Gumbel) shocks with scale $\sigma_{\text{help}} > 0$.

How commitment enters the parent budget (flow costs)

$$c = we(j, z) + (1 + r)a - a' - \text{housing} - \underbrace{\mathbf{1}\{\eta = 2\}\, \ell(\cdot)}_{\text{cosigning}} - \underbrace{\mathbf{1}\{\eta = 3\}\, \kappa(\cdot)}_{\text{cash}}.$$

Cosigning as expected liability (insurance-like stream)
The cosigning burden depends on the child's mortgage risk and size:

$$\ell(z^c, h_c) \;\text{ increases in leverage/house size } h_c \text{ and decreases in } z^c.$$

Child's portfolio is unobserved at commitment, so the parent internalizes the **expected liability**

$$\ell(\cdot) \equiv \mathbb{E}[\ell(z^c, h_c) \mid z^p, z^c].$$

Preferences Market Clearing Equilibrium

Calibration: Parameters

Table 12: Baseline Calibration

Parameter	Symbol	Value	Source
Preferences			
Discount factor	β	0.96	Various
Risk aversion	σ	1.5	Various
Housing share	θ	0.30	Garriga & Hedlund (2022)
Consumption–housing aggregator	ν	0.25	Garriga & Hedlund (2022)
Bequest strength	ϕ_1	-9.5	De Nardi (2004)
Bequest threshold	ϕ_2	11.6	De Nardi (2004)
Income process			
Persistence	ρ	0.96	Storesletten, Telmer & Yaron (2004)
Innovation s.d.	σ_ε	0.30	Various
Housing & prices			
Rental price	p^R	$0.06 \cdot p^O$	CBRE
Mortgage contract & borrowing			
Baseline LTV	λ_m	0.8	Fannie Mae
Maturity	M	30	Fannie Mae
Interest rate (risk-free)	r	0.03	Various

Targeted & Untargeted Moments

Table 13: Moments

	No Cosigning	With Cosigning	Data
Targeted			
Share Homeowner (Age $<$ 45)	0.287	0.58	0.44
Avg. Default Rate (Age $<$ 45)	0.004	0.002	0.035
Avg. Mortgage Interest Rate	0.068	0.685	0.061
Avg. Cosigning Rate		0.072	0.028
Untargeted			
Avg. PTI	0.288	0.197	0.44
Avg. Share Cash Transfer		0.02	0.08

Interpreting Cosigning through the model

How does intergenerational cosigning shape the housing distribution?
What are the consequences for intergenerational wealth transmission?

Table 14: Allowing Cosigning

	Δ Homeownership	Δ Avg. House Value	Δ Bequest
Low Type	-	-	-
Marginal Type	220%	-17.9%	1%
Rich Type	53%	-	-40%

Conclusion

Takeaways

- Cosigning expands mortgage access without upfront transfers.
- Correlations on the intensive margin driven by self–selection.
- A pricing reform in the secondary market reveals **heterogeneous contract adjustments** by cosigning status (rates vs. fees).

Quantitative model contribution

- Embeds irreversible parental support into a lifecycle housing model and quantifies effects on **homeownership, default risk, and wealth transmission**.

Next steps

- Natural experiment with web-scraped data.
- Refine calibration.
- Discipline model mechanisms using the reform-based moments and evaluate counterfactual policies affecting cosigning and affordability.

Appendix

Set Up

Modification of Allen et al. 2024 to dynamic setup.

$$U_1(c_1, h) = \ln(c_1) + \sigma \ln(h - h_0) \tag{1}$$

$$U_2(c_2, h) = \ln(c_2) + \sigma \ln(h - h_0) + \mu U_1^C(c_1^C, h^C)d \tag{2}$$

$$c_1 = y_1 - \phi ph - s \tag{3}$$

$$c_2 = y_2 - (1-\phi)(1+r_d)(1-d^P)ph - \kappa y_2 d + (1+r)s \tag{4}$$

where $d = (0, 1)$. Agents chose c_1, c_2, h, d.

$$h^* = \frac{\sigma\big[(1+\beta)(1+r)\beta y_1 + E(1-\kappa d)y_2\big] + H_0}{B(d^P) - A(d^P) + \sigma(1+r)\beta\phi} \tag{5}$$

$$PTI^* = \frac{(1-d^P)\sigma(1-\phi)(1+r_d)p\big[(1+\beta)(1+r)\beta y_1 + E(1-\kappa d)y_2\big] + H_0}{B(d^P) - A(d^P) + \sigma(1+r)\beta\phi} \tag{6}$$

with h^* increasing in d^P and PTI^* decreasing.

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Contrast to Cash Transfer

No help

$$\ln(y_1 - \phi p h^0 - s^0) + \sigma \ln(h^0 - h_0) +$$
$$\beta \Big[\ln(y_2 + (1+r)s^0) + \sigma \ln(h^0 - h_0) \Big]$$

Cosigning

$$\ln(y_1 - \phi p h^1 - s^1) + \sigma \ln(h^1 - h_0) +$$
$$\beta \Big[\ln(y_2 - \kappa y_2 + (1+r)s^1) + \sigma \ln(h^1 - h_0) + \mu U_1^C(c_1^c, h^c) \Big]$$

Cash Transfer

$$\ln(y_1 - \phi p h^2 - s^2 - \kappa y_1) + \sigma \ln(h^2 - h_0) +$$
$$\beta \Big[\ln(y_2 + (1+r)s^2) + \sigma \ln(h^2 - h_0) + \mu U_1^C(c_1^c, h^c) \Big]$$

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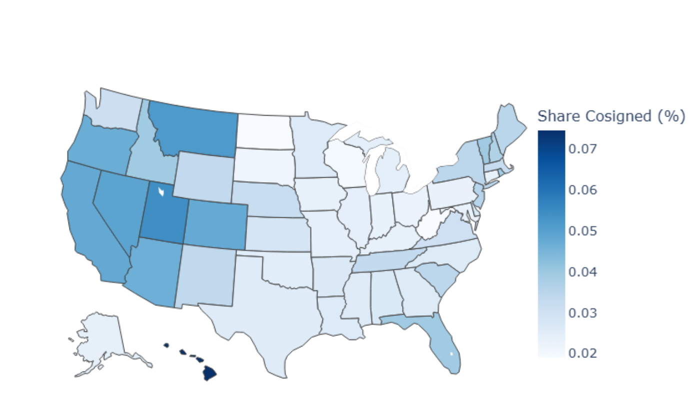
Variety across states

Figure 5: Share of Cosigned Mortgages, 2024



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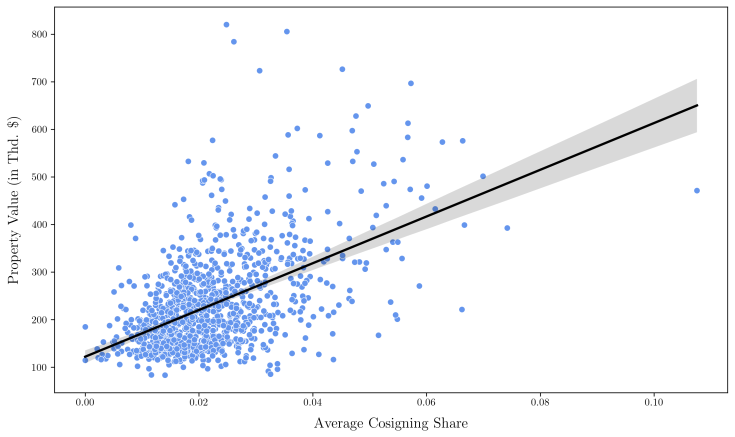
Property Prices - Larger Counties

Figure 6: Property Prices - At least 100 Mortgages per Year



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Extensive Margin Controls

Table 15: Extensive Margin & Cosigning - Adding Control

		Application Rejected	
Parental cosign	-0.0150***	-0.0434***	-0.0154**
	(0.0017)	(0.0019)	(0.0017)
County FE	No	Yes	No
Lender FE	Yes	No	Yes
State FE	No	No	No
Year FE	No	No	Yes
Demographics	Yes	Yes	Yes
Property Value & Loan Amount	Yes	Yes	Yes
R^2	0.009	0.086	0.009
Observations	6,672,649	6,672,649	6,672,649

Intensive Margin - Interest Rate

Table 16: Parental Cosigning on Mortgage Interest Rate

	Mortgage Interest Rate			
Parental cosigning	-0.0576***	-0.0964***	-0.0224**	-0.0415***
	(0.014)	(0.011)	(0.0110)	(0.0103)
County FE	No	Yes	No	Yes
Lender FE	Yes	No	Yes	Yes
State FE	No	No	No	No
Year FE	No	No	Yes	Yes
Demographics	Yes	Yes	Yes	Yes
R^2	0.001	0.012	0.000	0.000
Observations	5,437,803	5,437,803	5,437,803	5,437,803

Intensive Margin - Loan Amount

Table 17: Parental Cosigning on Loan Amount

Log Loan Amount				
Parental cosigning	0.0591***	0.0632***	0.0579***	0.0380***
	(0.0027)	(0.0027)	(0.0030)	(0.0030)
County FE	No	Yes	No	Yes
Lender FE	Yes	No	Yes	Yes
State FE	No	No	No	No
Year FE	No	No	Yes	Yes
Demographics	Yes	Yes	Yes	Yes
R^2	0.297	0.151	0.280	0.084
Observations	5,795,691	5,795,691	5,795,691	5,795,691

Intensive Margin - Property Value

Table 18: Parental Cosigning and Property Value

Log Property Value				
Parental cosigning	0.0900***	0.0985***	0.0889***	0.0681***
	(0.0027)	(0.0027)	(0.0027)	(0.0025)
County FE	No	Yes	No	Yes
Lender FE	Yes	No	Yes	Yes
State FE	No	No	No	No
Year FE	No	No	Yes	Yes
Demographics	Yes	Yes	Yes	Yes
R^2	0.324	0.163	0.304	0.093
Observations	5,795,691	5,795,691	5,795,691	5,795,691

Who are the children?

Table 19: Who are the children? - Some Means

	Savings	Total Wealth	Income**	Years Educ***
Not Cosigners	40,024	379,534	166,779	15.23
Cosigners	36,644	408,635	150,694	14.42

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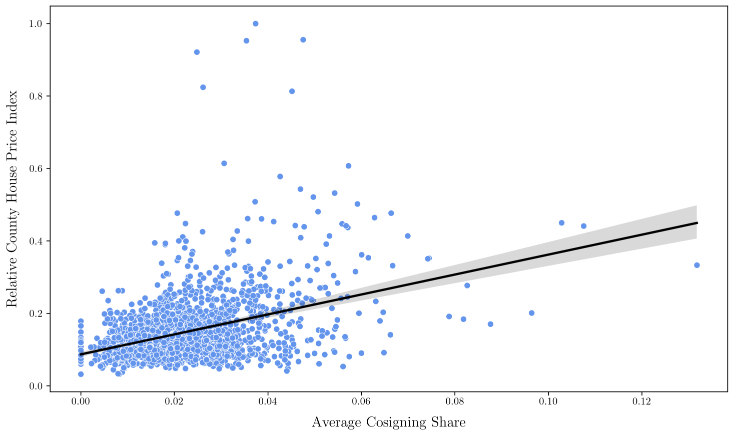
Figure 7: House Price Index



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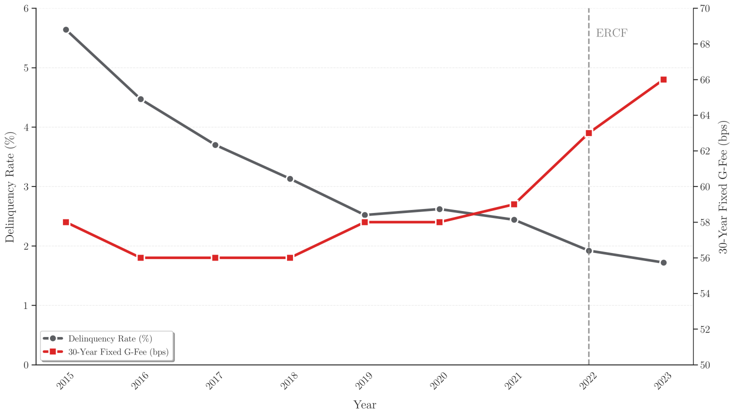
Guarantee Fees I

Figure 8: Guarantee Fees and Delinquency rates



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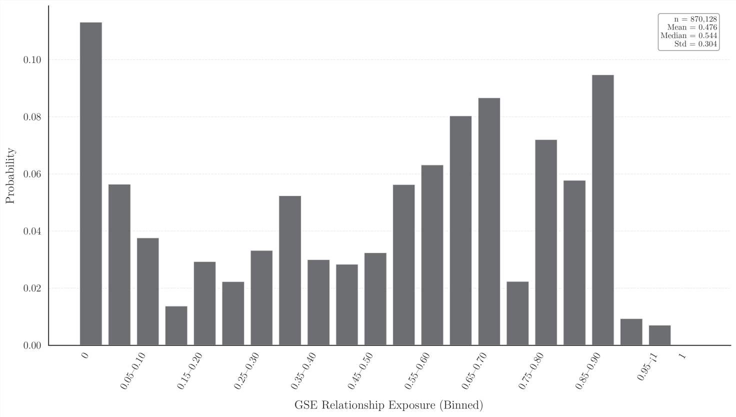
Exposure to GSE

Figure 9: GSE Exposure



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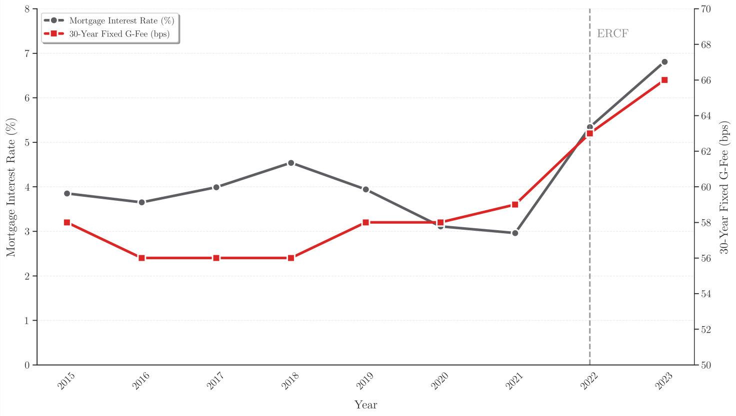
Guarantee Fees II

Figure 10: Guarantee Rates and Interest Rates



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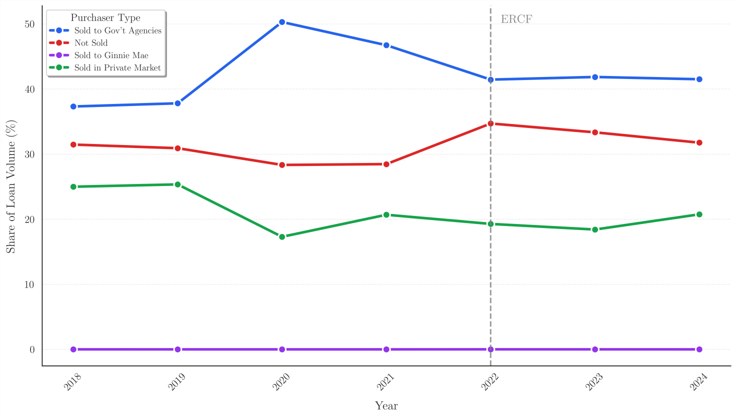
Loan Shares

Figure 11: Loan Shares



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Crucial Assumption: Time-Invariant Sorting Bias

Assumption 1 (Local Parallel Trends)

The discontinuity in potential outcomes absent the treatment is invariant across time.

$$\underbrace{B_1}_{\text{Post-Bias}} = \underbrace{B_0}_{\text{Pre-Bias}}$$

Formal Proposition:

$$\left(\lim_{x\downarrow 0} E[Y(0)|x, t=1] - \lim_{x\uparrow 0} E[Y(0)|x, t=1]\right)$$
$$= \left(\lim_{x\downarrow 0} E[Y(0)|x, t=0] - \lim_{x\uparrow 0} E[Y(0)|x, t=0]\right)$$

Implication: If the composition of borrowers at the threshold does not change differentially over time, B_0 is a valid counterfactual for B_1.

Robustness: Sensitivity to Bandwidth (50k–70k)

Table 20: Bandwidth Robustness

Outcome (β_3)	**(1)** **h = 50,000**	**(2)** **h = 60,000**	**(3)** **h = 70,000**
Panel A: Interest Rate (%)			
Non-Cosigner	-1.051***	-0.889***	-0.788***
Cosigner	-0.566***	-0.311*	-0.374**
Difference	**0.486****	**0.579*****	**0.414****
Panel B: Discount Points ($)			
Non-Cosigner	-319	-135	-52
Cosigner	878	1,527*	997
Difference	**1,197**	**1,662***	**1,049**
Panel C: Application Rejection			
Non-Cosigner	0.012	0.014*	0.010
Cosigner	-0.044	-0.060	-0.044
Difference	**-0.056**	**-0.074**	**-0.054**

Comparison of Diff-in-Disc estimates across fixed bandwidths.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

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Renter

Renting

$$V_{RB}^{rent}(s) = \max_{a',\, h^R \in \mathcal{H}^R} \left\{ u(c, h^R, 0) + \beta\, \mathbb{E}\left[V_{RB}(s') \right] \right\}$$

subject to

$$c = we(j, z) + (1 + r)a - a' - p^R h^R.$$

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Homeowner Stage: Pay vs. Default (with Extreme Value Shocks)

For $j \in \{j_B + 1, \ldots, j_B + M\}$, owners solve

$$V_{PD}(s) = \max\{V_{PD}^{pay}(s),\ V_{PD}^{def}(s)\}.$$

Pay

$$V_{PD}^{pay}(s) = \max_{a'} \left\{ u(c, h, 1) + \beta\, \mathbb{E}\left[V_{PD}(s')\right] + \varepsilon^{pay} \right\}$$

subject to

$$c = we(j, z) + (1 + r)a - a' - m(h, z).$$

Default

$$V_{PD}^{def}(s) = \max_{a'} \left\{ u(c, h, 1) - \xi + \beta\, \mathbb{E}\left[V_{\text{next}}(s')\right] + \varepsilon^{def} \right\}$$

subject to

$$c = we(j, z) + (1 + r)a - a'.$$

Extreme value shocks (logit)

$$\varepsilon^{pay}, \varepsilon^{def} \text{ i.i.d. Type-I Extreme Value (Gumbel).}$$

Then the conditional choice probabilities are

$$\exp\left(\frac{\tilde{V}_{PD}^{pay}(s)}{\sigma_{PD}}\right)$$

Preferences

Per-period utility (consumption and housing services)

$$u(c, h, \text{owner}) = \frac{\left(c^{1-\theta} \cdot (\chi(\text{owner})h)^{\theta}\right)^{1-\sigma} - 1}{1-\sigma}.$$

- $\chi(\text{owner}) > 1$ captures an ownership premium in housing services.
- Discount factor β; CRRA curvature σ; housing share θ.

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Terminal Period: Bequests and Intergenerational Wealth

At age $j = J$ the agent consumes and leaves bequests:

$$V(J, s) = u(c, h, \text{owner}) + b(a', h).$$

Terminal budget

$$c = we(J, z) + (1 + r)a - a' - \text{housing outlays}.$$

Bequest motive

$$b(a', h) = \phi_1 \cdot \frac{\left(a' + p^O h + \phi_2\right)^{1-\sigma} - 1}{1 - \sigma}.$$

Link to the paper: cosigning affects early-life housing access and leverage, shaping lifecycle portfolios and wealth transmission.

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Market Clearing and Equilibrium I

Prices are taken as given: (r, w, p^O, p^R) are exogenous.

State space

$$s \equiv (j, a, z, z^p, \omega, h, h^c, \eta),$$

where j is age, a liquid assets, z productivity, z^p parental type, $\omega \in \{RB, PD\}$ housing stage, h owned housing (if any), $h^c \in \{1, 2, 3\}$ help received (none/cosign/cash), and $\eta \in \{1, 2, 3\}$ irreversible help commitment made when old.

Optimal policy functions
Given prices and mortgage rules, the solution delivers:

$$a' = g_a(s), \quad h' = g_h(s), \quad \omega' = g_\omega(s), \quad d = g_d(s), \quad \eta' = g_\eta(s),$$

where $d \in \{0, 1\}$ is the default decision and $\eta' = \eta$ after commitment (irreversible).

1) Stationary distribution (consistency)
Let μ denote the cross-sectional distribution over s.

$$\mu' = \mathcal{T}(\mu; \{g_a, g_h, g_\omega, g_d, g_\eta\}, \Pi), \qquad \mu' = \mu.$$

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Market Clearing and Equilibrium II

2) Default consistency $\quad \forall\,(z^c, z^p, h)$

$$\overset{\text{guess}}{\Pr}\,(\text{default} \mid z^c, z^p, h) = \overset{\text{implied}}{\Pr}\,(\text{default} \mid z^c, z^p, h; \mu).$$

3) Cash-transfer consistency $\quad \forall\,(z^p, z^c, h)$

$$\pi^{\text{guess}}_{\text{cash},h}(z^p, z^c, h) = \pi^{\text{implied}}_{\text{cash},h}(z^p, z^c, h; \mu).$$

4) Cosigning consistency $\quad \forall\,(z^p, z^c, h)$

$$\pi^{\text{guess}}_{\text{cosign},h}(z^p, z^c, h) = \pi^{\text{implied}}_{\text{cosign},h}(z^p, z^c, h; \mu).$$

5) Inheritance consistency $\quad \forall\,(z^c, z^p)$

$$\text{inh}^{\text{guess}}(z^c, z^p) = \text{inh}^{\text{implied}}(z^c, z^p; \mu), \qquad \text{inh}^{\text{implied}}(z^c, z^p; \mu) = \bar{b}(z^p; \mu) \cdot \text{scale}(z^c, z^p).$$

Equilibrium: policy functions $\{g_a, g_h, g_\omega, g_d, g_\eta\}$ and a stationary distribution μ such that the implied conditional moments under μ coincide with the belief objects $\{\Pr(\text{default}), \pi_{\text{cash},h}, \pi_{\text{cosign},h}, \text{inh}\}$. Back

Extreme Value Shocks

Table 21: Baseline Calibration

Parameter	Symbol	Value	Source
Discrete-choice (EV shocks)			
Rent–buy logit scale	$\sigma_{\text{rent/buy}}$	0.5	
Help-type logit scale	σ_{help}	0.2	
Default logit scale	σ_{default}	0.2	
Default stigma	$-\xi$	7.0	

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